

Custom atoi() and atof() Implementation in C

1. Project Title : Implementation of atoi() and atof() Functions in C

2. Objective

The objective of this project is to implement the functionality of the standard C library functions atoi() and atof() without directly using the built-in conversion functions.

- The custom atoi() function converts a string into an integer.
- The custom atof() function converts a string into a floating-point number.
- The program handles positive and negative numbers.
- It processes the numeric portion of a string and stops when a non-numeric character is encountered.

3. Functions Implemented

A. Custom atoi() Function : The custom atoi() function converts the numeric portion of a string into an integer.

➤ General working

For example:

Input: "123"

Output: 123

For a negative number:

Input: "-123"

Output: -123

If numbers are followed by characters:

Input: "-123abc"

Output: -123

If the string does not begin with a valid number:

Input: "abc"

Output: 0

➤ Logic of atoi() : The function follows these steps:

1. Read the input string.
2. Check whether the first character is '-' or '+'.- 3. Store the sign accordingly.
- 4. Start reading the digits from the string.
- 5. For every digit:
 - Convert the character into its numerical value.
 - Add it to the resulting integer.
- 6. Stop when a non-digit character is encountered.
- 7. Apply the sign.
- 8. Return the final integer.

➤ Sample output of atoi() :

```

PS C:\Users\vishn\OneDrive\Documents\Desktop\VS> gcc atoi.c -o a1.exe
PS C:\Users\vishn\OneDrive\Documents\Desktop\VS> .\a1.exe
enter string:123
num=123
PS C:\Users\vishn\OneDrive\Documents\Desktop\VS> .\a1.exe
enter string:-123abc
num=-123
PS C:\Users\vishn\OneDrive\Documents\Desktop\VS> .\a1.exe
enter string:abc
num=0
PS C:\Users\vishn\OneDrive\Documents\Desktop\VS> █

```

B. Custom atof() Function : The custom atof() function converts a string into a floating-point number.

➤ **General working :**

For example:

Input: "23.5"

Output: 23.500000

For a negative floating-point number:

Input: "-46.5"

Output: -46.500000

It also handles a numeric value followed by characters:

Input: "-123abc"

Output: -123.000000

➤ **Logic of atof() :** The function performs the following steps:

1. Read the input string.
2. Check for a positive or negative sign.
3. Read the digits before the decimal point.
4. Construct the integer part.
5. Check for the decimal point '.'.
6. If a decimal point exists, process the digits after it.
7. Convert the fractional digits into their corresponding decimal values.
8. Add the fractional part to the integer part.
9. Apply the sign.
10. Return the final floating-point value.

➤ **Sample output of atof() :**

```

Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\vishn\OneDrive\Documents\Desktop\VS> gcc atof.c -o a.exe
PS C:\Users\vishn\OneDrive\Documents\Desktop\VS> .\a.exe
Enter string:23.5
num=23.500000
PS C:\Users\vishn\OneDrive\Documents\Desktop\VS> .\a.exe
Enter string:-46.5
num=-46.500000
PS C:\Users\vishn\OneDrive\Documents\Desktop\VS> .\a.exe
Enter string:-123abc
num=-123.000000
PS C:\Users\vishn\OneDrive\Documents\Desktop\VS> █

```